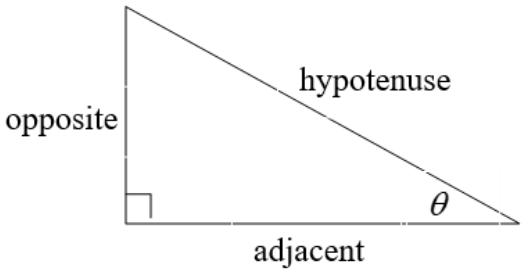
## Definition of the Trig Functions

### Right triangle definition

For this definition we assume that $0 < \theta < \frac{\pi}{2}$ or $0^\circ < \theta < 90^\circ$.



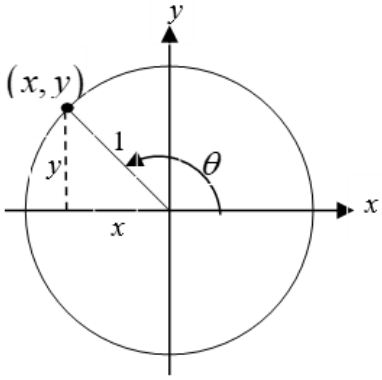

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \csc(\theta) = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \sec(\theta) = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} \qquad \cot(\theta) = \frac{\text{adjacent}}{\text{opposite}}$$

### Unit Circle Definition

For this definition $\theta$ is any angle.




$$\sin(\theta) = \frac{y}{1} = y \qquad \csc(\theta) = \frac{1}{y}$$

$$\cos(\theta) = \frac{x}{1} = x \qquad \sec(\theta) = \frac{1}{x}$$

$$\tan(\theta) = \frac{y}{x} \qquad \cot(\theta) = \frac{x}{y}$$

## Facts and Properties

### Domain

The domain is all the values of $\theta$ that can be plugged into the function.

$\sin(\theta)$, $\theta$ can be any angle

$\cos(\theta)$, $\theta$ can be any angle

$\tan(\theta)$, $\theta \neq \left(n + \frac{1}{2}\right)\pi,\ n = 0, \pm1, \pm2, \ldots$

$\csc(\theta)$, $\theta \neq n\pi,\ n = 0,\ \pm1,\ \pm2, \ldots$

$\sec(\theta)$, $\theta \neq \left(n + \frac{1}{2}\right)\pi,\ n = 0, \pm1, \pm2, \ldots$

$\cot(\theta)$, $\theta \neq n\pi,\ n = 0, \pm1, \pm2, \ldots$

### Period

The period of a function is the number, $T$, such that $f(\theta + T) = f(\theta)$. So, if $\omega$ is a fixed number and $\theta$ is any angle we have the following periods.

$$\sin(\omega\,\theta) \quad \rightarrow \quad T = \frac{2\pi}{\omega}$$

$$\cos(\omega\,\theta) \quad \rightarrow \quad T = \frac{2\pi}{\omega}$$

$$\tan(\omega\,\theta) \quad \rightarrow \quad T = \frac{\pi}{\omega}$$

$$\csc(\omega\,\theta) \quad \rightarrow \quad T = \frac{2\pi}{\omega}$$

$$\sec(\omega\,\theta) \quad \rightarrow \quad T = \frac{2\pi}{\omega}$$

$$\cot(\omega\,\theta) \quad \rightarrow \quad T = \frac{\pi}{\omega}$$

### Range

The range is all possible values to get out of the function.

$$-1 \le \sin(\theta) \le 1 \qquad -1 \le \cos(\theta) \le 1$$

$$-\infty < \tan(\theta) < \infty \qquad -\infty < \cot(\theta) < \infty$$

$$\sec(\theta) \ge 1 \text{ and } \sec(\theta) \le -1 \qquad \csc(\theta) \ge 1 \text{ and } \csc(\theta) \le -1$$

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# Formulas and Identities

### Tangent and Cotangent Identities

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} \qquad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

### Reciprocal Identities

$$\csc(\theta) = \frac{1}{\sin(\theta)} \qquad \sin(\theta) = \frac{1}{\csc(\theta)}$$
$$\sec(\theta) = \frac{1}{\cos(\theta)} \qquad \cos(\theta) = \frac{1}{\sec(\theta)}$$
$$\cot(\theta) = \frac{1}{\tan(\theta)} \qquad \tan(\theta) = \frac{1}{\cot(\theta)}$$

### Pythagorean Identities

$$\sin^2(\theta) + \cos^2(\theta) = 1$$
$$\tan^2(\theta) + 1 = \sec^2(\theta)$$
$$1 + \cot^2(\theta) = \csc^2(\theta)$$

### Even/Odd Formulas

$$\sin(-\theta) = -\sin(\theta) \qquad \csc(-\theta) = -\csc(\theta)$$
$$\cos(-\theta) = \cos(\theta) \qquad \sec(-\theta) = \sec(\theta)$$
$$\tan(-\theta) = -\tan(\theta) \qquad \cot(-\theta) = -\cot(\theta)$$

### Periodic Formulas

If $n$ is an integer then,

$$\sin(\theta + 2\pi n) = \sin(\theta) \qquad \csc(\theta + 2\pi n) = \csc(\theta)$$
$$\cos(\theta + 2\pi n) = \cos(\theta) \qquad \sec(\theta + 2\pi n) = \sec(\theta)$$
$$\tan(\theta + \pi n) = \tan(\theta) \qquad \cot(\theta + \pi n) = \cot(\theta)$$

### Degrees to Radians Formulas

If $x$ is an angle in degrees and $t$ is an angle in radians then

$$\frac{\pi}{180} = \frac{t}{x} \quad \Rightarrow \quad t = \frac{\pi x}{180} \quad \text{and} \quad x = \frac{180t}{\pi}$$

### Double Angle Formulas

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$
$$\begin{aligned}\cos(2\theta) &= \cos^2(\theta) - \sin^2(\theta)\\ &= 2\cos^2(\theta) - 1\\ &= 1 - 2\sin^2(\theta)\end{aligned}$$
$$\tan(2\theta) = \frac{2\tan(\theta)}{1 - \tan^2(\theta)}$$

### Half Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{2}}$$
$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos(\theta)}{2}}$$
$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}}$$

### Half Angle Formulas (alternate form)

$$\sin^2(\theta) = \tfrac{1}{2}(1 - \cos(2\theta)) \qquad \tan^2(\theta) = \frac{1-\cos(2\theta)}{1+\cos(2\theta)}$$
$$\cos^2(\theta) = \tfrac{1}{2}(1 + \cos(2\theta))$$

### Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$$
$$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$$
$$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha)\tan(\beta)}$$

### Product to Sum Formulas

$$\sin(\alpha)\sin(\beta) = \tfrac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$$
$$\cos(\alpha)\cos(\beta) = \tfrac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$
$$\sin(\alpha)\cos(\beta) = \tfrac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$
$$\cos(\alpha)\sin(\beta) = \tfrac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

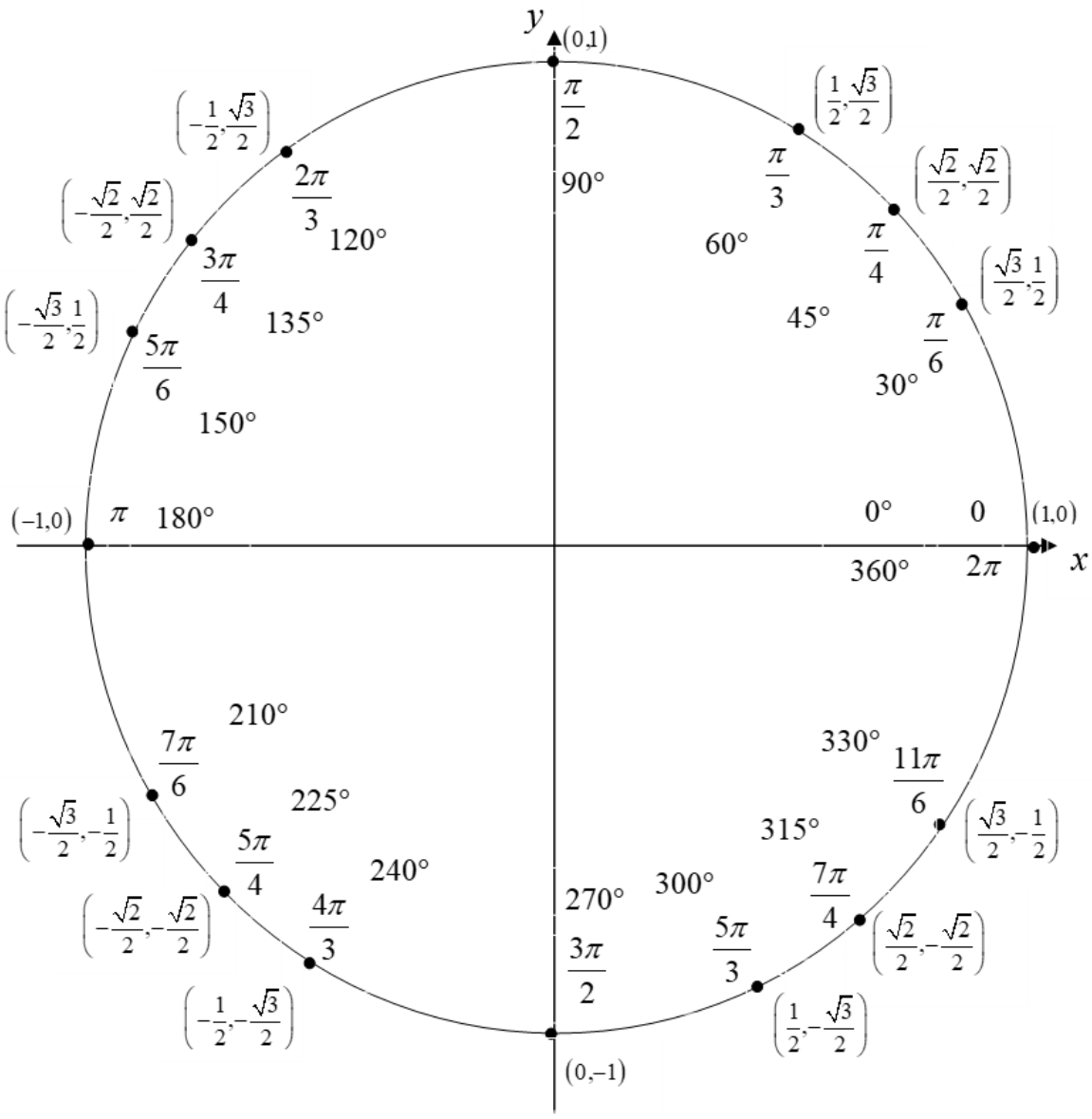
### Sum to Product Formulas

$$\sin(\alpha) + \sin(\beta) = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\sin(\alpha) - \sin(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$
$$\cos(\alpha) + \cos(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

### Cofunction Formulas

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta) \qquad \cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$$
$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec(\theta) \qquad \sec\left(\frac{\pi}{2} - \theta\right) = \csc(\theta)$$
$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta) \qquad \cot\left(\frac{\pi}{2} - \theta\right) = \tan(\theta)$$

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For any ordered pair on the unit circle $(x, y)$ : $\cos(\theta) = x$ and $\sin(\theta) = y$

**Example**

$$\cos\left(\frac{5\pi}{3}\right) = \frac{1}{2} \qquad \sin\left(\frac{5\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

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# Inverse Trig Functions

### Definition

$y = \sin^{-1}(x)$ is equivalent to $x = \sin(y)$

$y = \cos^{-1}(x)$ is equivalent to $x = \cos(y)$

$y = \tan^{-1}(x)$ is equivalent to $x = \tan(y)$

### Domain and Range

| Function | Domain | Range |
|---|---|---|
| $y = \sin^{-1}(x)$ | $-1 \le x \le 1$ | $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ |
| $y = \cos^{-1}(x)$ | $-1 \le x \le 1$ | $0 \le y \le \pi$ |
| $y = \tan^{-1}(x)$ | $-\infty < x < \infty$ | $-\frac{\pi}{2} < y < \frac{\pi}{2}$ |

### Inverse Properties

$\cos\left(\cos^{-1}(x)\right) = x$ $\quad \cos^{-1}\left(\cos(\theta)\right) = \theta$

$\sin\left(\sin^{-1}(x)\right) = x$ $\quad \sin^{-1}\left(\sin(\theta)\right) = \theta$

$\tan\left(\tan^{-1}(x)\right) = x$ $\quad \tan^{-1}\left(\tan(\theta)\right) = \theta$

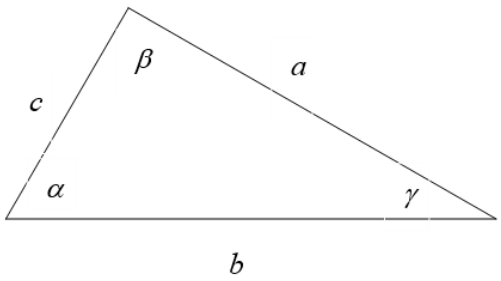
### Alternate Notation

$\sin^{-1}(x) = \arcsin(x)$

$\cos^{-1}(x) = \arccos(x)$

$\tan^{-1}(x) = \arctan(x)$

# Law of Sines, Cosines and Tangents




### Law of Sines

$$\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$$

### Law of Cosines

$a^2 = b^2 + c^2 - 2bc\cos(\alpha)$

$b^2 = a^2 + c^2 - 2ac\cos(\beta)$

$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$

### Mollweide's Formula

$$\frac{a+b}{c} = \frac{\cos\left(\frac{1}{2}(\alpha-\beta)\right)}{\sin\left(\frac{1}{2}\gamma\right)}$$

### Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{1}{2}(\alpha-\beta)\right)}{\tan\left(\frac{1}{2}(\alpha+\beta)\right)}$$

$$\frac{b-c}{b+c} = \frac{\tan\left(\frac{1}{2}(\beta-\gamma)\right)}{\tan\left(\frac{1}{2}(\beta+\gamma)\right)}$$

$$\frac{a-c}{a+c} = \frac{\tan\left(\frac{1}{2}(\alpha-\gamma)\right)}{\tan\left(\frac{1}{2}(\alpha+\gamma)\right)}$$

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